Hospital Emergency Room Dashboard Analysis

Interactive Excel Dashboard using Power Query, Power Pivot & DAX

Executive Summary

Efficient emergency room management is essential for improving patient care, minimizing wait times, and optimizing hospital resources. This project presents an interactive Hospital Emergency Room Dashboard developed in Microsoft Excel. The dashboard transforms raw hospital data into meaningful visual insights that help monitor patient flow, service efficiency, admissions, referrals, and satisfaction levels.

The dashboard enables healthcare managers to evaluate operational performance through interactive filters, KPI cards, and dynamic visualizations.

Project Overview

The Hospital Emergency Room Dashboard was developed to analyze patient records and visualize important operational metrics. Using Microsoft Excel's Power Query, Power Pivot, DAX, Pivot Tables, and Pivot Charts, the dashboard provides an interactive view of emergency room performance.

Users can filter the dashboard by month and year to evaluate trends and identify operational bottlenecks.

Business Objectives

The dashboard was designed to answer the following questions:

- How many patients visited the emergency room?
- What is the average patient wait time?
- How satisfied are patients with the service?
- What percentage of patients were attended within the target time?
- What percentage of patients required admission?
- Which departments receive the highest referrals?
- Which age groups visit the emergency room most frequently?

Dataset Description

The dataset contains emergency room patient records including:

- Patient ID
- Visit Date
- Age
- Gender
- Admission Status
- Wait Time
- Satisfaction Score
- Department Referral
- Attendance Status
- Month
- Year

Tools & Technologies

- Microsoft Excel
- Power Query
- Power Pivot
- DAX
- Pivot Tables
- Pivot Charts
- Slicers
- Conditional Formatting

Data Preparation

The dataset was cleaned using Power Query.

Cleaning activities included:

- Removing duplicate records
- Handling missing values
- Standardizing column names
- Formatting dates
- Creating Month and Year columns
- Preparing the data model

Data Modeling

Power Pivot was used to establish relationships and support DAX calculations.

Calculated fields were created to enable:

- Patient counts
- Wait time analysis
- Satisfaction score
- Admission statistics
- Attendance performance

Dashboard Overview

The final dashboard provides an interactive and comprehensive overview of **Hospital Emergency Room performance** by consolidating key operational metrics into a single, user-friendly interface. It features KPI cards, interactive charts, and slicers that enable users to analyze patient volume, average wait time, patient satisfaction, admission status, attendance performance, demographic distribution, and department referrals. The dashboard supports dynamic filtering by month and year, allowing stakeholders to monitor trends, evaluate operational efficiency, and make informed, data-driven decisions.

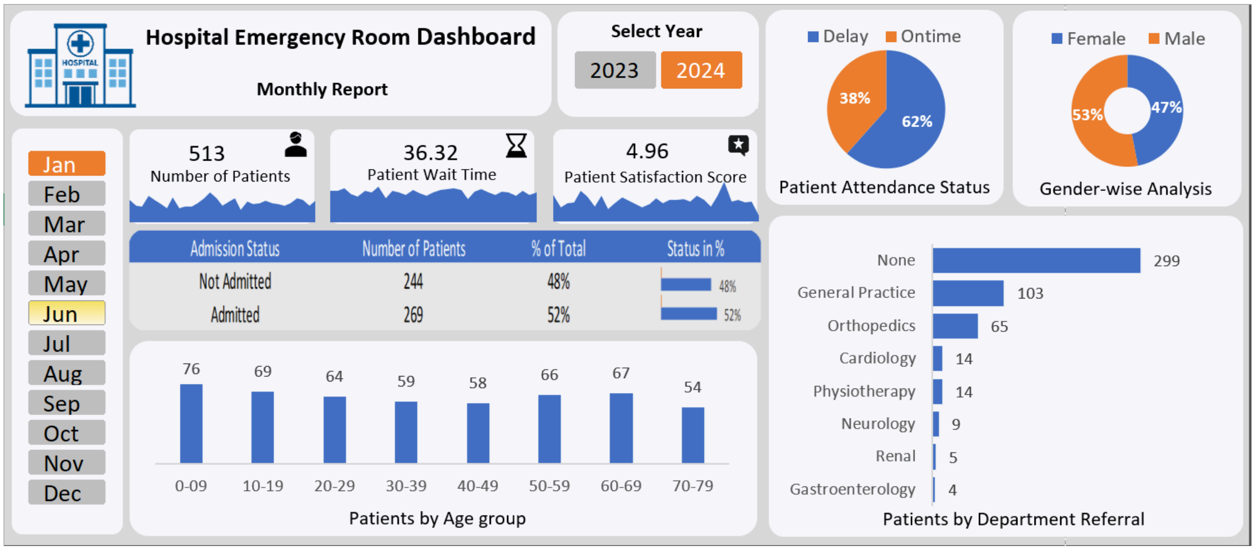
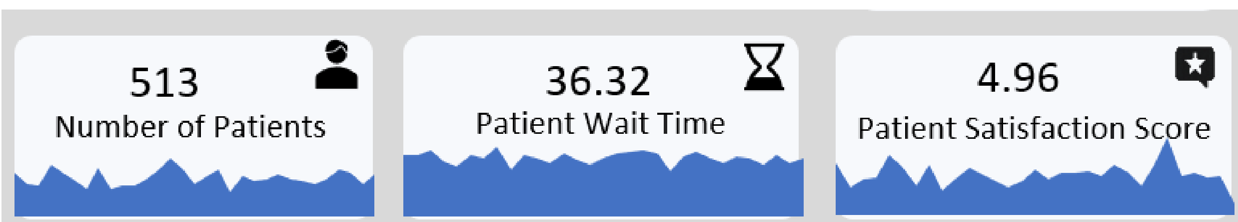


Figure 1. Hospital Emergency Room Dashboard

Dashboard KPIs



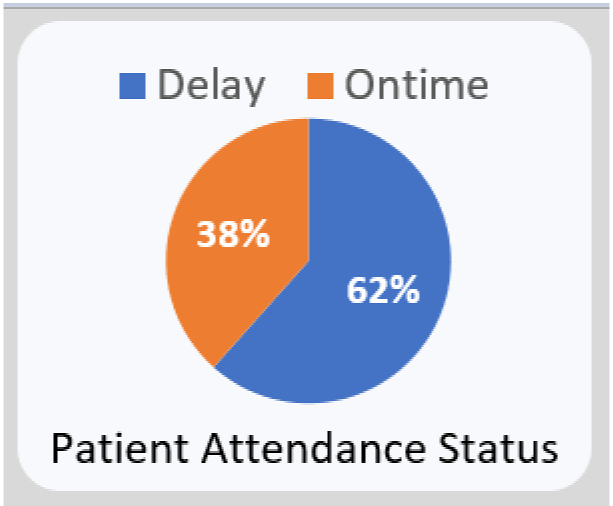
The dashboard displays the following KPIs:

KPI	Value
Total Patients	**513**
Average Wait Time	**36.32 Minutes**
Patient Satisfaction Score	**4.96 / 10**

Dashboard Visualizations

1. Patient Attendance Status

Compares patients attended on time against delayed cases.



Result:

- On Time: **38%**
- Delay: **62%**

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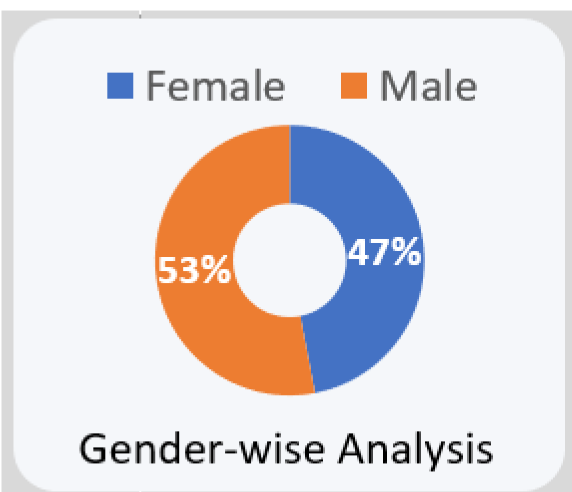
2. Gender-wise Analysis



Shows patient distribution by gender.

- Female: **47%**
- Male: **53%**

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3. Admission Status

Displays admitted versus non-admitted patients.

Admission Status	Number of Patients	% of Total	Status in %
Not Admitted	244	48%	48%
Admitted	269	52%	52%

- Admitted: **269**
- Not Admitted: **244**

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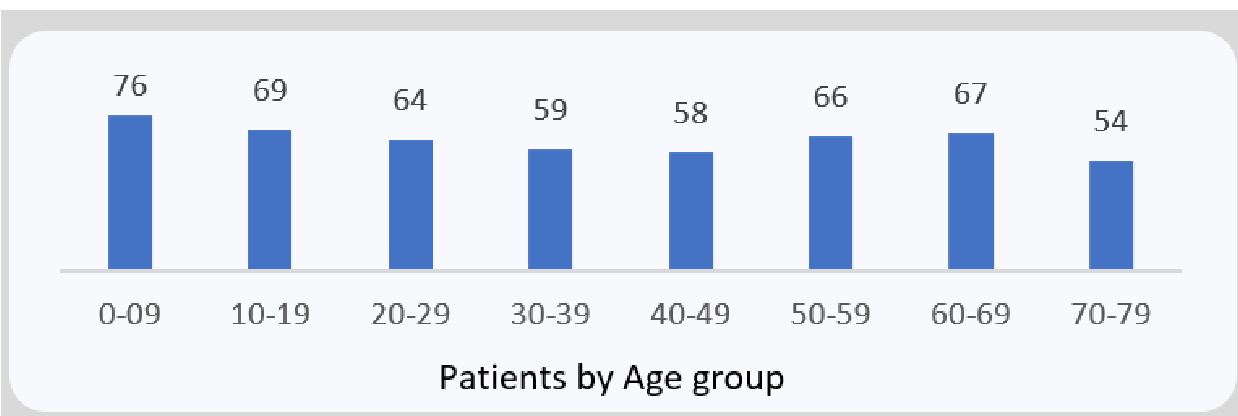
4. Patient Age Group

Analyzes emergency room visits across age categories.



The highest patient volume comes from the younger and middle-age groups, helping identify which demographics most frequently use emergency services.

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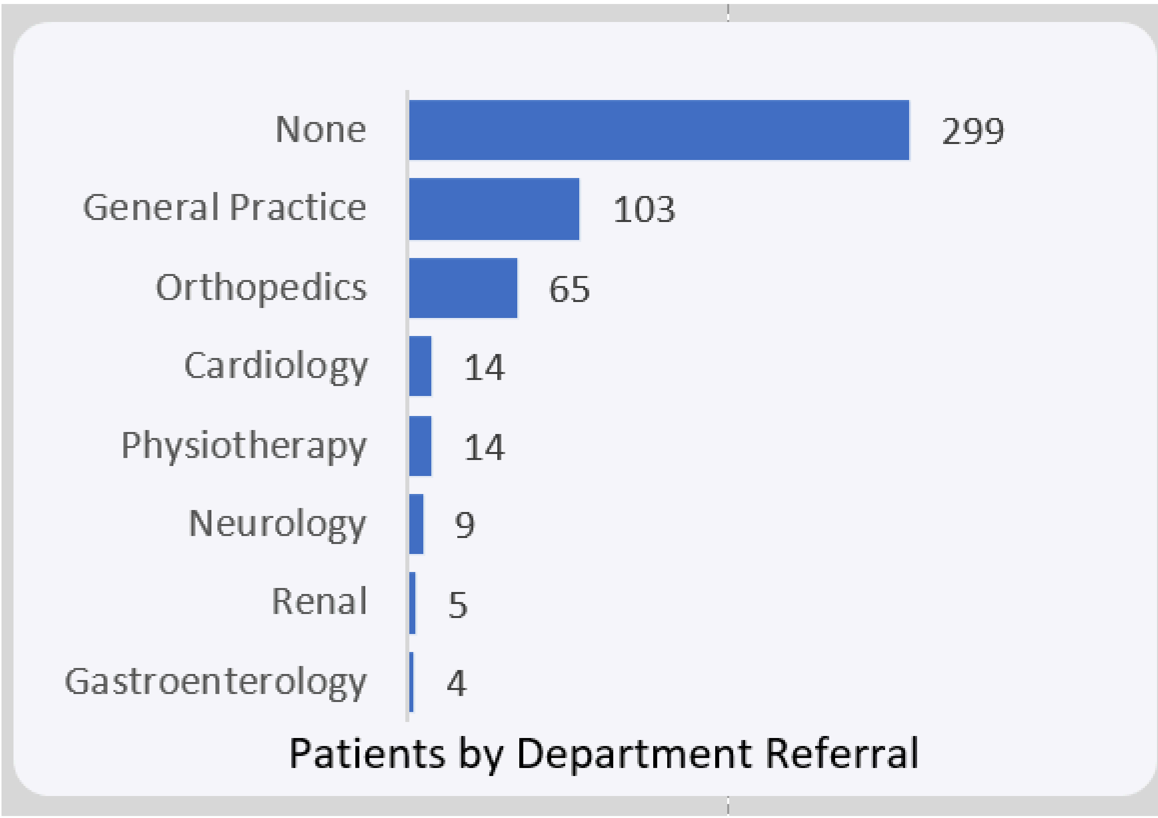
5. Department Referral Analysis

Ranks departments based on referral frequency.



Top referrals include:

- None
- General Practice
- Orthopedics
- Cardiology
- Physiotherapy

Key Insights

- The emergency room handled **513 patient visits** during the selected reporting period.
- **38%** of patients were attended within the expected service time, while **62%** experienced delays.
- The **average waiting time** was **36.32 minutes**, indicating opportunities to improve operational efficiency.
- Patient satisfaction averaged **4.96**, suggesting room to enhance the overall patient experience.
- **Male patients (53%)** slightly outnumbered female patients (47%).
- Nearly half of all patients (**52%**) required hospital admission.

- Most referrals were categorized as **None**, indicating that many cases were resolved within the emergency department without specialist consultation.
- **General Practice** and **Orthopedics** were the most common referral departments.

Business Recommendations

- Reduce patient waiting times by improving staff scheduling during peak hours.
- Investigate factors contributing to lower patient satisfaction.
- Monitor delayed attendance rates regularly.
- Allocate additional resources to frequently referred departments.
- Use age-group trends for better workforce planning and resource allocation.

Skills Demonstrated

- Microsoft Excel Dashboard Development
- Power Query
- Power Pivot
- DAX
- Data Cleaning
- Data Modeling
- KPI Reporting
- Healthcare Analytics
- Data Visualization
- Business Intelligence

Conclusion

The Hospital Emergency Room Dashboard demonstrates how Microsoft Excel can be used to transform healthcare data into an interactive business intelligence solution. By integrating Power Query, Power Pivot, DAX, and dynamic visualizations, the dashboard enables stakeholders to monitor emergency room performance, identify operational challenges, and support data-driven decision-making.

Prepared By

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